

Hamud Abdul Aziz

South Jakarta, Indonesia

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SUMMARY

An undergraduate Computer Science student with a strong interest in data analysis and intelligent systems. Experienced in working with large data problems and building problem-solving models. Eager to leverage analytical skills and a detail-oriented mindset to help derive actionable insights from data in a professional team.

EDUCATION

2023 - Present Bachelor of Computer Science at **Universitas Padjadjaran** (GPA: 3.80/4.0)
2020 - 2023 Multimedia at **SMKN 59 Jakarta** (Grades: 90.98)

EXPERIENCES

Department of Computer Science Unpad Feb 2025 – June 2026

Laboratory Teaching Assistant

- Supported laboratory instruction across four undergraduate courses: Database Systems, Algorithms & Programming, OOPs, and Computer Networking.
- Mentored and assessed 150+ student across two years with the average grades of 83+.

Center for Innovation in Teaching and Learning (PIPP) Unpad Oct 2025 - Jan 2026

Asset Developer Intern

- Pioneered and played a key role in the comprehensive virtualization of the campus, successfully recreating the university's environment and atmosphere within the Roblox engine.
- Modeled 9+ iconic university buildings with 1:1 precision and developed 2 interactive faculty-themed mini-games to enhance user engagement and provide an immersive

ORGANIZATION EXPERIENCES

RDAT, BEM Kema FMIPA Unpad May 2025 – Jan 2026

Backend Lead & Co-Lead Consultant Research Team

- Served as Backend Lead for the BEM FMIPA website, responsible for API development, database design, and system architecture decisions.
- Appointed as Co-Vice Consultant for Research, supporting research project development, methodology structuring, and analytical review within the bureau and the student organization as a whole.

SELECTED PROJECTS

Automated Garment Tracking & Quality Control System Apr 2026 – On Going

Full-Stack Developer

- Engineered a real-time production tracking system using Next.js and FastAPI, enabling factory admins to monitor daily garment batches and worker activities seamlessly.
- Developed an integrated QR Code scanning module for field workers to perform automated scan-in/scan-out, reducing manual logging errors.

NutriMatch: AI-Powered Personalized Diet & Allergen-Smart Recommendation System Apr 2026 – Jun 2026

AI Engineer, Coding Camp 2026 powered by DBS Foundation

- Analyzed and processed multi-dimensional user health data (age, gender, BMI, activity levels, and diet goals) to build automated calculators for BMR, TDEE, and daily caloric targets.
- Developed data-driven insight engines by engineering custom training pipelines and wrapping the predictive models into live endpoints, enabling automated, personalized 7-day meal plan recommendations.

Backend Developer

- Designed and developed a secure, scalable RESTful API architecture for an e-commerce and ecological crowdfunding web application using Node.js/Express and PostgreSQL.
- Developed an AI-driven product review analysis system using a Decision Tree model and built its backend endpoint to process textual feedback parameters, enabling automated sentiment and rating classification.

Network Architect & Designer

- Designed high-speed end-to-end network topologies to support massive video data transmissions and secure cross-city collaboration between Jakarta and Bandung studios.
- Established secure inter-branch WAN connections by configuring IPsec VPN over GRE Tunnels with AES and SHA-1 encryption over a simulated public ISP infrastructure.

3D Animator & UI Designer

- Conceptualized and developed 8 distinct medical-response animations using Blender to realistically simulate real-time clinical patient symptoms and trauma reactions.
- Designed lifelike character rigging and visual asset feedback loops for critical scenarios, including choking indicators, respiratory distress reactions, and post-injection recoveries.

Data Analyst

- Engineered an end-to-end machine learning system to process 16 years of NASA satellite data, resolving severe data gaps using linear interpolation and climatology baselines to ensure reliable analysis.
- Trained and evaluated multiple regression algorithms for temperature forecasting, achieving a highly accurate Mean Absolute Error of 0.645°C and an R² score of 0.7560 utilizing a Random Forest model.
- Deployed the forecasting model and unsupervised anomaly detection algorithms (Isolation Forest, Local Outlier Factor) into a real-time monitoring interface to successfully track and isolate high-confidence subsurface thermal events.

SKILLS

Languages	English(Fluent), Bahasa Indonesia(Native)
Programming Languages	C++, C#, Python, Javascript, PHP (CodeIgniter, Laravel)
Tools and Libraries	VSCode, Git, Unity, MySQL, Figma.